

Inductor Design Problems

1.

- a. The waveforms in Figure 2a. show the current flowing in an inductor and the voltage across the inductor.

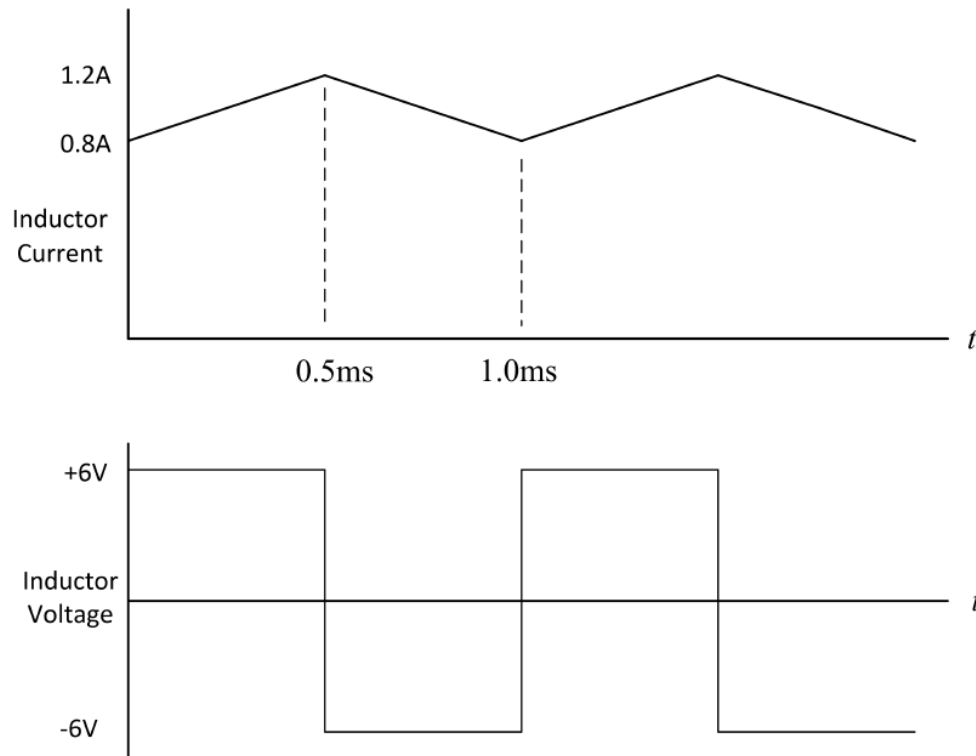


Figure 2a. Inductor Current and Voltage Waveforms

- b. You need to design an inductor for this application. You plan to use a ferrite RM8 core, with an effective cross sectional area of 63 mm^2 , and a maximum flux density of around 0.3 tesla. Calculate the necessary inductance, how many turns are required for the inductor, and the necessary air gap.

2.

- b) Magnetic materials used for inductors in power electronic circuits have three important characteristics commonly referred to as permeability, saturation, and hysteresis. With reference to what is going on inside the magnetic material, explain what each of these characteristics are.